COMMERCIAL ITEM DESCRIPTION

BEADING MACHINE, SHEET METAL, ROTARY, MANUALLY OPERATED

The General Services Administration has authorized the use of this commercial item decryption in preference to type I of military specification MIL-F-23243A.

1. SCOPE

1.1 Scope. This description covers manually operated, rotary, sheet metal beading machines of the standard commercial heavy duty type which are used in a production environment.

1.2 Classification. Beading machines shall be of one type and of the following sizes. The size to be furnished shall be as specified (see 6.1).

Size 18 - 18 gauge (0.0478 inch) capacity.
Size 20 - 20 gauge (0.0359 inch) capacity.
Size 22 - 22 gauge (0.0299 inch) capacity.
Size 24 - 24 gauge (0.0239 inch) capacity.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on the date of invitation for bids or requests for proposal, form a part of this description to the extent specified herein.

Military Specification MIL-M-18058 -	Machinery, Metal and Woodworking, Packaging of.
Federal Standard FED-STD-376 -	Preferred Metric Units for General Use by the Federal Government.
American Society for Testing and Materials, ASTM D 3951 -	Standard Practice for Commercial Packaging.

3. SALIENT CHARACTERISTICS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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3.1 Design. Beading machines shall be new and of the manufacturer's standard design in effect on the date of the solicitation. The machines shall be capable of performing beading operations on cylindrical or flat workplaces of sheet metal of the rated thickness capacity specified in table I for the size machine required. The machines shall meet the requirements specified herein and shall conform to current commercial Industrial practice.

TABLE I. Beading machine requirements.

Requirements	Size			
	18	20	22	24
Capacity mild steel, inches	0.0478	0.0359	0.0299	0.0239
Distance between shaft centers, inches (min.)	2-2/3	4-1/8	2-1/3	1-7/8
Distance, center of rolls to gauge, inches (min.)	12-3/4	29-1/4	7-1/3	6-1/4

NOTE: Requirements shall be not less than value specified.

3.2 Frame. The frame shall be an iron or steel casting, capable of supporting shafts and drive mechanism, providing a suitable base connection for mounting.

3.3 Stand. Each beading machine shall be provided with a stand for bench mounting. The size 18 machines shall have an integral cast frame and stand with the base having four holes for mounting to a bench or floor stand. Sizes 20, 22, and 24 shall be provided with a clamp-on offset stand for bench mounting with the offset bringing the machine toward the operator.

3.4 Shafts. The shafts shall be machine finished steel. Shafts shall be driven-hand crank and gears and shall be fitted to bearings.

3.4.1 Bearings. The bearings shall conform to standard commercial bearings used in manually operated beading machines furnished to the industrial market.

3.5 Gears. All gears shall be cast iron or machine cut steel and shall be totally enclosed or guarded to protect the operator.

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3.6 Rolls. The rolls shall be machine finished steel and shall be of sufficient hardness to satisfactorily perform the operations for which designed. Unless otherwise specified (see 6.1), the following beading rolls shall be furnished:

a. Sizes 18, 20, and 22 machines.
   (1) One pair of 3/8 inch single beading rolls.
   (2) One pair of 3/4 inch ogee (OG) beading rolls.

b. Size 24 machines.
   (1) One pair of 1/4 inch single beading rolls.
   (2) One pair of 3/4 inch OG beading rolls.

3.7 Hand crank. The hand crank shall be of sufficient length to allow operation of the beading machine without undue strain to the operator.

3.8 Adjustment device. The upper roll shall be-adjustable by means of a quick acting crankscrew for controlling the roll position in relation to the workpiece.

3.8.1 Gauge. The machine shall be provided with an adjustable gauge to govern the bead placement on the workpiece. The gauge shall be of smooth finish and shall be fitted to the machine.

3.9 Measurement system. All dimensions in this description are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.10 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government’s policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier’s employees to undue hazards from the recovered materials.

3.11 Workmanship. The workmanship of the beading machines shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order. the contractor is responsible for the performance of all “inspect Ion, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Examination. Each beading maachine shall be visually examined to determine conformance with the requirements of this description.

4.3 Operational inspection. Each beading machine shall be operated without performing beading operations to ensure freedom of movement of the gears, adjusting mechanisms, and Interchangeability of rolls.

4.4 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with. MIL-M-18058.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.
b. Size of beading machine required (see 1.2).
c. Specify rolls required, if different (see 3.6).
d. Packaging in acccordance with MIL-M-18058, if required specify level of preservation and packing (see 5.1).

6.2 Cross reference data. Items covered by Military Specification MIL-F-23243A, type are cross referenced to items covered by this description as follows:

MIL-F-23243A	A-A-51159
Type I	(Deleted) Size 18, 20, 22, and 24 (New size)

MILITARY INTERESTS:

Custodians

Army - AL
Air Force - 99

Review activities

Navy - YD
Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3441-0116

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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